SCSI1113: Computational Mathematics

~~2015-2016/ Semester 2~~ Sem 1 Sesi 2021/2022

~~**TUTORIAL 4**~~ Assignment 6 Eigen and Interpolation

Question 1

Given, $A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ -4 & 8 & -4 & 2 \\ -7 & 7 & -3 & 3 \\ -2 & 2 & -2 & 4 \end{bmatrix}$

a) Use the power method to find the dominant eigenvalue. Let $v^{(0)}=(1,0,0,0)^T$ and iterate until $\varepsilon=0.005$.

b) Use the shifted power method to find the eigenvalue of the smallest magnitude. Let $v^{(0)}=(0,0,1,0)^T$ and iterate until $\varepsilon=0.005$.

Question 2

Matrix $\boldsymbol{B}$ is given as follows, [20 marks]

$$B = \begin{bmatrix} -18 & 20 & -6 \\ -12 & 14 & -3 \\ 30 & -30 & 13 \end{bmatrix}$$

a) Use the Gerschgorin's Circle Theorem to determine a region containing all the eigenvalues of $\boldsymbol{B}$.

b) Find the dominant eigenvalue (λ_1) and the corresponding eigenvector of matrix $\boldsymbol{B}$ using Power method. Use $\mathbf{v}^{(0)} = [0, 0, 1]^T$.

c) Find the smallest eigenvalue (λ_3) and the corresponding eigenvector of matrix $\boldsymbol{B}$ using Shifted Power method. Use $\mathbf{v}^{(0)} = [0, 0, 1]^T$.

d) Find the intermediate eigenvalue (λ_2) of matrix $\boldsymbol{B}$.

Note: Do calculation in 4 decimal points with $\varepsilon=0.05$. Round up the dominant eigenvalue (λ_1) in (b) and the smallest eigenvalue (λ_3) in (c) to the nearest integer.

Question 3

Using the following data for the heat capacity of methylcyclohexane as a function of temperature:

Temperature	150	200	250	300
Capacity	1.43	1.54	1.70	1.89

Interpolate to estimate the heat capacity at:

a) 175 using Newton forward-difference formula

b) 225 using Newton divided-difference formula

c) 275 using Newton backward-difference formula

Question 4

Given the following data:

x	-1	-0.75	-0.25	0.25	0.75	1
$f(x)$	0	-0.7	-0.7	0.7	0.7	0

a) Determine the approximation value of $f(0.5)$ and $f(0)$ based on Lagrance interpolating polynomials method.

b) Compare the approximation value of $f(0.5)$ and $f(0)$ with the corresponding values of $f(x) = \sin(\pi x)$.

Question 5

Find the quadratic least squares approximation to the following data.

x	-4	-3	-2	-1	0	1	2	3	4	5
y	8.0	9.9	1.8	1.4	1.7	3.6	7.5	16.8	14.1	27.2

Approximate $y(1.8)$ and $y(-2.3)$.